

An exchange move for cyclically decreasing factorisations, and the reduction of WZZ Problem 5.1 to a single hypothesis

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Abstract

Wang, Zhang and Zhang [4] prove that the affine Stanley symmetric polynomial $\tilde{F}_w(x_1, \dots, x_r)$ has M-convex support, by way of Lam’s theorem that $\tilde{F}_w$ expands positively in dual k -Schur functions [3, Cor. 8.5]; their Problem 5.1 asks for a direct proof from the cyclically-decreasing-decomposition definition. We give an explicit *exchange move* on cyclically decreasing factorisations into two factors (Theorem 4.1), proved from the definition by a block-rotation model of cyclically decreasing elements, and consuming neither [3, Cor. 8.5] nor any property of dual k -Schur functions. Two consequences follow. First, the set of achievable lengths of one factor in a chain, with its two neighbours held fixed, depends only on the quotient of the neighbours and is a *full integer interval* (Theorem 3.1, Corollary 4.4); in particular the “Markov” hypothesis of the cylindric argument of [1] collapses here to a statement about a single group element. Second, Problem 5.1 reduces to the single remaining hypothesis that the support has a dominance maximum (Theorem 5.5). We show by an explicit counterexample (Proposition 5.6) that this last hypothesis does *not* follow from the interval and palindrome properties, so it is a genuine gap and not an oversight; we state it precisely and record the computational evidence for it.

1 Introduction

1.1 The problem

Let $n = k + 1 \geq 2$ and let $\tilde{S}_n$ be the affine symmetric group. Following [4, (def-affine-stanley)], an affine permutation $v \in \tilde{S}_n$ is *cyclically decreasing* if it has a reduced word $s_{i_1} \cdots s_{i_\ell}$ in which all generators are distinct and s_{i+1} precedes s_i whenever both occur; a *cyclically decreasing decomposition* of w is a factorisation $w = w^1 w^2 \cdots w^r$ with each w^t cyclically decreasing and $\ell(w) = \sum_t \ell(w^t)$; and

$$\tilde{F}_w(x_1, \dots, x_r) = \sum_{w=w^1 \cdots w^r} x_1^{\ell(w^1)} \cdots x_r^{\ell(w^r)}, \quad (1)$$

the sum being over all cyclically decreasing decompositions into r factors.

[4, Thm. 4.7] states that $\tilde{F}_w(x_1, \dots, x_r)$ is M-convex, i.e. that

$$W_r(w) := \sup \tilde{F}_w(x_1, \dots, x_r) = \{(\ell(w^1), \dots, \ell(w^r))\} \subseteq \mathbb{Z}^r$$

satisfies the symmetric exchange axiom. Their proof routes through [3, Cor. 8.5], which they call a deep result, and Problem 5.1 asks for a *direct* proof from (1). The success criterion is therefore not the truth of M-convexity — that is already a theorem — but the provenance of the argument. Every statement below is graded accordingly, and §6 lists every external input.

1.2 What we prove

Write $A = \mathbb{Z}/n$ and, for a proper subset $S \subsetneq A$, let u_S denote the unique cyclically decreasing element with support S (Lemma 2.2); $\ell(u_S) = |S|$. For $v \in \tilde{S}_n$ put

$$\mathcal{F}(v) = \{ |S| : v = u_S u_T \text{ with } \ell(v) = |S| + |T|, S, T \subsetneq A \}.$$

- **Theorem 3.1 (localisation).** If $w = w^1 \cdots w^r$ is a cyclically decreasing decomposition, $1 \leq t \leq r-1$, $p = w^1 \cdots w^{t-1}$ and $q = w^1 \cdots w^{t+1}$, then the set of values of $\ell(w^t)$ obtainable while holding every other factor fixed is exactly $\mathcal{F}(p^{-1}q)$. It depends on p and q only through $v = p^{-1}q$.
- **Theorem 4.1 (exchange move).** If $v = u_S u_T$ is length-additive and $|S| > |T|$, then $v = u_{S'} u_{T'}$ is length-additive for an explicit $S' \subset S$ with $|S'| = |S| - 1$ and $T' = T \cup \{m\}$. The move is: choose a maximal cyclic run $[m, M]$ of S with $m \notin T$, let e be the first element of $m, m+1, \dots, M$ with $e+1 \notin T$, and set $S' = S \setminus \{e\}$, $T' = T \cup \{m\}$.
- **Corollary 4.4.** $\mathcal{F}(v)$ is a full integer interval.
- **Theorem 5.5 (reduction).** Granting that $\tilde{F}_w$ is a symmetric function [2] — a fact independent of [3, Cor. 8.5] — the set $W_r(w)$ is S_r -stable and its set of sorted vectors is a dominance order ideal. Consequently WZZ Problem 5.1 follows from *Hypothesis* (H4) alone: that this ideal has a dominance maximum.
- **Proposition 5.6.** (H4) is independent of everything proved here: there is a set $W \subseteq \mathbb{Z}^3$ that is homogeneous, S_3 -stable, and has palindromic full interval fibres in every pair of coordinates, yet is not M-convex.

1.3 Relation to the cylindric argument

In [1] the author proved M-convexity of cylindric skew Schur polynomials from a template with two stated hypotheses: (H1) the weight is the increment sequence of a scalar statistic, and (H2) a chain step conditioned on both neighbours ranges over an integer box. Re-reading §§3–6 of that paper for the present transfer shows that the template in fact has *four* hypotheses. The palindrome identity [1, Lem. 3.2] is used twice — once to build the cylindric Bender–Knuth involution and once inside the exchange corollary — and is not a consequence of (H1) and (H2); call it (H3). The existence of a chain maximising every prefix of the statistic [1, Lem. 5.2(3)] is a fourth, (H4). Proposition 5.6 shows (H4) is not implied by (H1)–(H3), so the omission was not harmless.

For affine Stanley polynomials, (H1) is free with the statistic ℓ , and Theorem 3.1 shows that (H2) is not a Markov hypothesis at all: the fibre depends only on $p^{-1}q$. Theorem 4.1 then supplies (H2), and (H3) follows from the symmetry of $\tilde{F}_w$. Only (H4) remains.

2 The affine symmetric group and the block model

Definition 2.1. $\tilde{S}_n$ is the group of bijections $w : \mathbb{Z} \rightarrow \mathbb{Z}$ with $w(i+n) = w(i) + n$ for all i and $\sum_{i=1}^n w(i) = \binom{n+1}{2}$. For $i \in \mathbb{Z}/n$ let $s_i \in \tilde{S}_n$ be the transposition interchanging $i+jn$ and $i+1+jn$ for every $j \in \mathbb{Z}$. Products act by $(ab)(j) = a(b(j))$, so right multiplication permutes positions: $(ws_i)(j) = w(s_i(j))$. One has $\ell(ws_i) = \ell(w) + 1$ if and only if $w(i) < w(i+1)$.

Throughout, a subset $S \subseteq \mathbb{Z}/n$ is identified with its n -periodic lift $\tilde{S} = \{j \in \mathbb{Z} : j \bmod n \in S\}$; we write S for both. A *run* of S is a maximal interval $[m, M] \subseteq \mathbb{Z}$ contained in S . If S is proper then its complement $G = \mathbb{Z} \setminus S$ (the *gap set*) is nonempty and n -periodic, so all runs are finite. The *blocks* of S are the intervals $[g + 1, g']$ where $g < g'$ are consecutive gaps; thus each run $[m, M]$ gives the block $[m, M + 1]$, and a pair of adjacent gaps $g, g + 1$ gives the singleton block $[g + 1, g + 1]$. The blocks partition $\mathbb{Z}$.

Lemma 2.2 (Block model). *Let $S \subsetneq \mathbb{Z}/n$. There is a unique cyclically decreasing element with support S , namely the permutation u_S that rotates each block downwards by one step:*

$$u_S(g + 1) = g', \quad u_S(j) = j - 1 \quad (g + 1 < j \leq g') \quad (2)$$

for every block $[g + 1, g']$. Equivalently $u_S(j) = j - 1$ if $j - 1 \in S$, and $u_S(j) = \min\{c \geq j : c \notin S\}$ if $j - 1 \notin S$. Moreover $\ell(u_S) = |S|$.

Proof. Let $[m, M]$ be a run of S . The product $s_M s_{M-1} \cdots s_m$ is a reduced word in distinct generators in which s_{i+1} precedes s_i whenever both occur; as a permutation it sends $m \mapsto M + 1$ and $j \mapsto j - 1$ for $m < j \leq M + 1$, i.e. it is the rotation (2) of the block $[m, M + 1]$ and the identity elsewhere. Distinct runs $[m, M], [m', M']$ are separated by at least one gap, so the blocks $[m, M + 1], [m', M' + 1]$ are disjoint and the corresponding rotations commute; their product is the permutation described, and its length is $\sum (M - m + 1) = |S|$ because the concatenated word is reduced. (Reducedness is the case $w = \text{id}$ of Lemma 2.3, whose proof multiplies out the word $s_M s_{M-1} \cdots s_m$ letter by letter using only the criterion “ $\ell(ws_i) > \ell(w)$ iff $w(i) < w(i + 1)$ ” and therefore does not depend on the present lemma; for $w = \text{id}$ the criterion reads $M + 1 > j$ for $m \leq j \leq M$, which holds.) Any cyclically decreasing element with support S has a reduced word using each s_i , $i \in S$, exactly once with s_{i+1} before s_i whenever both occur; since S is proper, each run must be read from its top down, and letters from different runs commute, so the element is u_S . Uniqueness follows. $\square$

Lemma 2.3 (Length-additivity criterion). *Let $w \in \tilde{S}_n$ and $S \subsetneq \mathbb{Z}/n$. Then $\ell(wu_S) = \ell(w) + |S|$ if and only if for every run $[m, M]$ of S ,*

$$w(M + 1) > w(j) \quad \text{for all } j \in [m, M].$$

Proof. For a single run, $u_{[m, M]} = s_M s_{M-1} \cdots s_m$. Multiplying on the right one letter at a time, $\ell(ws_M) = \ell(w) + 1$ iff $w(M) < w(M + 1)$; the element ws_M has $M + 1$ and M swapped in position, so $\ell(ws_M s_{M-1})$ increases iff $(ws_M)(M - 1) < (ws_M)(M)$, i.e. iff $w(M - 1) < w(M + 1)$; inductively the j -th step increases the length iff $w(M - j + 1) < w(M + 1)$. Hence the run contributes $M - m + 1$ to the length iff $w(M + 1) > w(j)$ for all $j \in [m, M]$. Distinct runs act on disjoint position-blocks and commute, and the condition for each involves only w ; so the conditions are simultaneous. $\square$

3 Localisation of the fibre

Theorem 3.1 (Localisation). *Let $w = w^1 \cdots w^r$ be a cyclically decreasing decomposition, let $1 \leq t \leq r - 1$, and put $p = w^1 \cdots w^{t-1}$, $q = w^1 \cdots w^{t+1}$, $v = p^{-1}q$. Then*

$$\{\ell(x) - \ell(p) : p \cdots x \cdots q \text{ replaces } w^t, w^{t+1} \text{ by a cyclically decreasing pair}\} = \mathcal{F}(v).$$

In particular the fibre depends on p and q only through v , and not at all on $w^1, \dots, w^{t-1}, w^{t+2}, \dots, w^r$.

Proof. The original decomposition is length-additive, so $\ell(q) = \ell(p) + \ell(v)$. Suppose $v = u_S u_T$ with $\ell(v) = |S| + |T|$. Then

$$\ell(p) + \ell(v) = \ell(q) = \ell(pu_S u_T) \leq \ell(pu_S) + \ell(u_T) \leq \ell(p) + |S| + |T| = \ell(p) + \ell(v),$$

so equality holds throughout: $\ell(pu_S) = \ell(p) + |S|$ and $\ell(q) = \ell(pu_S) + |T|$. Hence $x = pu_S$ is an admissible replacement with $\ell(x) - \ell(p) = |S|$, and replacing (w^t, w^{t+1}) by (u_S, u_T) leaves the whole decomposition length-additive. Conversely any admissible x gives $u_S = p^{-1}x$ and $u_T = x^{-1}q$ cyclically decreasing with $|S| + |T| = \ell(v)$. The factors outside positions $t, t+1$ are untouched throughout. $\square$

Remark 3.2. This is where the affine arena is genuinely simpler than the cylindric one. In [1] hypothesis (H2) is a Markov property: the admissible middle row is constrained by both neighbours and the constraint does not factor. Here the two-sided constraint collapses, by the subadditivity squeeze above, to a one-sided statement about the single element $v = p^{-1}q$.

4 The exchange move

Theorem 4.1 (Exchange move). *Let $v = u_S u_T$ with $\ell(v) = |S| + |T|$ and $|S| > |T|$. Then there exist a run $[m, M]$ of S with $m \notin T$ and an $e \in [m, M]$ with $e+1 \notin T$; for the first such e in the order $m, m+1, \dots, M$, putting*

$$S' = S \setminus \{e\}, \quad T' = T \cup \{m\},$$

one has $u_{S'} u_{T'} = v$ and $\ell(v) = |S'| + |T'|$, with $|S'| = |S| - 1$.

We prove the two halves separately.

Lemma 4.2 (Applicability). *Let $v = u_S u_T$ with $\ell(v) = |S| + |T|$. Call a run $[m, M]$ of S usable if $m \notin T$ and $j+1 \notin T$ for some $j \in [m, M]$. If no run of S is usable, then $|T| \geq |S|$.*

Proof. Step 1: if m is the bottom of a run $[m, M]$ of S and $m \in T$, then $[m, M+1] \subseteq T$. Let $[a, b]$ be the run of T containing m ; it is finite because T is proper. Apply Lemma 2.3 with $w = u_S$ to the run $[a, b]$ of T (legitimate since $\ell(u_S u_T) = \ell(u_S) + |T|$): we get $u_S(b+1) > u_S(j)$ for all $j \in [a, b]$, in particular for $j = m$. Now m is the bottom of the S -block $[m, M+1]$, so $u_S(m) = M+1$ by (2). Suppose $b \leq M$. If $b = m$ then $b+1 = m+1 \in [m+1, M+1]$, so $u_S(b+1) = m$, and $m > u_S(m) = M+1$ is false. If $m < b \leq M$ then $b+1 \in [m+1, M+1]$, so $u_S(b+1) = b \leq M < M+1 = u_S(m)$, again false. Hence $b \geq M+1$, and since $a \leq m$ we get $[m, M+1] \subseteq [a, b] \subseteq T$.

Step 2: counting. Let $R_1, \dots, R_\rho$ be the runs of S , one per period, $R_i = [m_i, M_i]$ and $L_i = M_i - m_i + 1$, so that $\sum_i L_i = |S|$. Assume no run is usable and set

$$A = \{i : m_i \in T\}, \quad B = \{1, \dots, \rho\} \setminus A.$$

For $i \in B$ we have $m_i \notin T$ and, since R_i is not usable, $j+1 \in T$ for every $j \in R_i$; put $X_i = [m_i + 1, M_i + 1]$, of size L_i . For $i \in A$, Step 1 gives $[m_i, M_i + 1] \subseteq T$; put $X_i = [m_i, M_i + 1]$, of size $L_i + 1$. In both cases $X_i \subseteq T$ and

$$X_i \subseteq R_i \cup \{M_i + 1\}.$$

Read modulo n the sets X_i are pairwise disjoint: the runs R_i are pairwise disjoint; $M_j + 1 \notin S$ while $R_i \subseteq S$, so $M_j + 1 \notin R_i$; and $M_i + 1 = M_j + 1$ forces $i = j$. Each X_i also injects into $\mathbb{Z}/n$, since $|X_i| \leq L_i + 1 \leq n$, and $|X_i| = n$ would force $T = \mathbb{Z}/n$, contradicting properness. Therefore

$$|T| \geq \sum_{i \in A} (L_i + 1) + \sum_{i \in B} L_i = |S| + |A| \geq |S|. \quad \square$$

Lemma 4.3 (The identity). *Let $v = u_S u_T$ with $\ell(v) = |S| + |T|$ and $|T| \leq n - 2$. Let $[m, M]$ be a run of S with $m \notin T$, and let e be the first element of $m, m + 1, \dots, M$ with $e + 1 \notin T$ (assumed to exist). Then $u_{S \setminus \{e\}} u_{T \cup \{m\}} = v$, and the factorisation is length-additive.*

Proof. Since $|T| \leq n - 2$, the gap set of T has at least two elements per period, so consecutive gaps of T differ by less than n . As $m \notin T$, m is a gap of T ; let c be the previous gap, so $0 < m - c < n$. By minimality of e we have $m + 1, \dots, e \in T$ and $e + 1 \notin T$, so $e + 1$ is the next gap of T after m . Thus the two T -blocks meeting at m are $[c + 1, m]$ and $[m + 1, e + 1]$.

Write $T' = T \cup \{m\}$. Its gap set is that of T with the orbit of m removed, so the two blocks above merge into the single block $[c + 1, e + 1]$; all other blocks are unchanged. Comparing (2) for T and for T' , the two permutations agree everywhere except

$$u_T(c + 1) = m, \quad u_{T'}(c + 1) = e + 1, \quad u_T(m + 1) = e + 1, \quad u_{T'}(m + 1) = m.$$

Hence $u_{T'} = u_T t$, where t is the n -periodic involution interchanging $c + 1 + jn$ and $m + 1 + jn$ for all j ; it is well defined because $0 < (m + 1) - (c + 1) < n$.

Write $S' = S \setminus \{e\}$. Its gap set is that of S with the orbit of e added, so the S -block $[m, M + 1]$ containing e splits into $[m, e]$ and $[e + 1, M + 1]$, all other blocks being unchanged. Comparing (2) for S and for S' ,

$$u_S(m) = M + 1, \quad u_{S'}(m) = e, \quad u_S(e + 1) = e, \quad u_{S'}(e + 1) = M + 1,$$

and the two agree elsewhere. Hence $u_{S'} = u_S t'$, where t' is the n -periodic involution interchanging $m + jn$ and $e + 1 + jn$; it is well defined because $0 < (e + 1) - m \leq M + 1 - m \leq n - 1$.

Now conjugate: $u_T t u_T^{-1}$ is the involution interchanging $u_T(c + 1)$ and $u_T(m + 1)$, that is, interchanging m and $e + 1$. So $u_T t u_T^{-1} = t'$, and therefore

$$u_{S'} u_{T'} = u_S t' u_T t = u_S (u_T t u_T^{-1}) u_T t = u_S u_T t^2 = u_S u_T = v.$$

Finally T' is proper because $|T'| = |T| + 1 \leq n - 1$, and $|S'| + |T'| = (|S| - 1) + (|T| + 1) = \ell(v)$; since $\ell(v) \leq \ell(u_{S'}) + \ell(u_{T'}) = |S'| + |T'|$ always, equality holds and the factorisation is length-additive. $\square$

Proof of Theorem 4.1. By Lemma 4.2 and $|S| > |T|$, a usable run $[m, M]$ exists. Also $|T| < |S| \leq n - 1$ gives $|T| \leq n - 2$, so Lemma 4.3 applies and yields the factorisation. $\square$

Corollary 4.4. *For every $v \in \tilde{S}_n$, $\mathcal{F}(v)$ is a (possibly empty) set of consecutive integers.*

Proof. Let $\ell = \ell(v)$. Let $\Psi(z) = \phi(z^{-1})$, where ϕ is the automorphism of $\tilde{S}_n$ induced by $s_i \mapsto s_{-i}$. Then Ψ is a length-preserving anti-automorphism, and $\Psi(u_S) = u_{-S}$: indeed u_S^{-1} is the reverse word, which is cyclically *increasing* with support S , and negating all indices turns a cyclically increasing word on S into a cyclically decreasing word on $-S$. Hence $v = u_S u_T$ is length-additive iff $\Psi(v) = u_{-T} u_{-S}$ is, so

$$\mathcal{F}(\Psi(v)) = \ell - \mathcal{F}(v) := \{\ell - k : k \in \mathcal{F}(v)\}. \quad (3)$$

Theorem 4.1 says: $k \in \mathcal{F}(v)$ and $k > \ell - k$ imply $k - 1 \in \mathcal{F}(v)$. Applying this to $\Psi(v)$ and using (3) gives the dual statement: $k \in \mathcal{F}(v)$ and $k < \ell - k$ imply $k + 1 \in \mathcal{F}(v)$. So every $k \in \mathcal{F}(v)$ is connected inside $\mathcal{F}(v)$ to $\lfloor \ell/2 \rfloor$ or to $\lceil \ell/2 \rceil$ by unit steps; as $\lfloor \ell/2 \rfloor$ and $\lceil \ell/2 \rceil$ are equal or adjacent, $\mathcal{F}(v)$ is an interval. $\square$

Corollary 4.5 ((H2) for affine Stanley). *In the situation of Theorem 3.1, the set of achievable values of $\ell(w^t)$ with all other factors held fixed is a full integer interval containing the original value.*

5 Consequences, and the single remaining hypothesis

We use one external fact.

Hypothesis 5.1 (Symmetry; [2], quoted at [4, 1.796, 1.802]). $\tilde{F}_w$ is a symmetric function. Equivalently, the coefficient of $x_1^i x_2^j$ in $\tilde{F}_v(x_1, x_2)$ equals that of $x_1^j x_2^i$.

This is Lam's original 2006 result (the commutation of the cyclically decreasing generating elements in the affine nilCoxeter algebra), *not* [3, Cor. 8.5], and so is admissible under the success criterion of Problem 5.1. See §6 and §8 for its status.

Corollary 5.2 (Palindromicity). $\min \mathcal{F}(v) + \max \mathcal{F}(v) = \ell(v)$ whenever $\mathcal{F}(v) \neq \emptyset$.

Proof. $\tilde{F}_v(x_1, x_2) = \sum_{k \in \mathcal{F}(v)} c_k x_1^k x_2^{\ell(v)-k}$ with $c_k > 0$. By Hypothesis 5.1 the multiset of exponents is invariant under $k \mapsto \ell(v) - k$, so $\mathcal{F}(v)$ is. $\square$

Proposition 5.3 (S_r -stability and the dominance ideal). $W_r(w)$ is stable under S_r , and the set $P(W_r(w)) = \{\text{sort}(\alpha) : \alpha \in W_r(w)\}$ of sorted weight vectors is an order ideal of the dominance order on partitions of $\ell(w)$ with at most r parts.

Proof. Let $\alpha \in W_r(w)$ and $1 \leq t \leq r - 1$; let v be as in Theorem 3.1 and $N = \alpha_t + \alpha_{t+1} = \ell(v)$. By Corollary 4.4 and Corollary 5.2, $\mathcal{F}(v) = [\mu_-, \mu_+]$ with $\mu_- + \mu_+ = N$ and $\alpha_t \in [\mu_-, \mu_+]$. Since $N - \alpha_t \in [\mu_-, \mu_+]$ as well, the vector obtained from α by interchanging α_t and α_{t+1} lies in $W_r(w)$; adjacent transpositions generate S_r , giving stability. If moreover $\alpha_t > \alpha_{t+1}$, then $2\mu_- \leq \mu_- + \mu_+ = N < 2\alpha_t$, so $\alpha_t - 1 \geq \mu_-$ and $\alpha - e_t + e_{t+1} \in W_r(w)$.

Now let $\nu \in P(W_r(w))$ and $\sigma \leq \nu$. By the Robin Hood step [1, Lem. 4.1] — if $\nu \triangleright \sigma$ there are $a < b$ with $\nu_a \geq \nu_b + 2$ such that $\tau = \nu - e_a + e_b$ is a partition with $\nu \triangleright \tau \geq \sigma$ — and induction on the number of steps, it suffices to treat $\sigma = \nu - e_a + e_b$. By S_r -stability $W_r(w)$ contains a vector β whose entries are those of ν with $\beta_t = \nu_a > \nu_b = \beta_{t+1}$ for some t ; the previous paragraph gives $\beta - e_t + e_{t+1} \in W_r(w)$, whose sorted form is σ . $\square$

Hypothesis 5.4 ((H4), dominance maximum). For every $w \in \tilde{S}_n$ and $r \geq 1$ with $W_r(w) \neq \emptyset$, the ideal $P(W_r(w))$ has a dominance maximum $\hat{\lambda}$; equivalently, some cyclically decreasing decomposition maximises $\ell(w^1 \cdots w^t)$ simultaneously for every t .

Theorem 5.5 (Reduction). *Assume Hypothesis 5.1 and Hypothesis 5.4. Then $W_r(w)$ is M -convex for all w and r , and*

$$\text{Newton } \tilde{F}_w(x_1, \dots, x_r) = \mathcal{P}_{\hat{\lambda}},$$

the permutohedron of $\hat{\lambda}$. In particular WZZ Problem 5.1 holds, with an argument that uses neither [3, Cor. 8.5] nor any property of dual k -Schur functions.

Proof. By Proposition 5.3, $P(W_r(w))$ is a dominance ideal, and it is S_r -stable as a subset of $\mathbb{Z}^r$; by Hypothesis 5.4 it has a maximum $\hat{\lambda}$. A dominance ideal with a maximum, viewed as an S_r -stable subset of $\mathbb{Z}^r$, is M-convex and equals $\mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^r$: this is [1, §6], which is self-contained polymatroid theory proved from the exchange axiom and cites nothing. $\square$

Proposition 5.6 ((H4) is independent). *Let $W \subseteq \mathbb{Z}_{\geq 0}^3$ be the union of the S_3 -orbits of $(4, 1, 1)$, $(3, 3, 0)$, $(3, 2, 1)$ and $(2, 2, 2)$, a set of 13 vectors. Then*

1. *every $\alpha \in W$ has $\alpha_1 + \alpha_2 + \alpha_3 = 6$;*
2. *for every $\alpha \in W$ and every pair $i \neq j$, the set $\{\beta_i : \beta \in W, \beta_s = \alpha_s \ (s \neq i, j)\}$ is a set of consecutive integers whose minimum and maximum sum to $\alpha_i + \alpha_j$;*
3. *W is not M-convex.*

Hence the conclusions of Corollaries 4.4 and 5.2 and Proposition 5.3 do not imply M-convexity, and Hypothesis 5.4 is a genuine additional input.

Proof. (1) is clear. (2) and (3) are finite verifications; (3) is witnessed by $\alpha = (1, 1, 4)$ and $\beta = (0, 3, 3)$ with $i = 1$: we have $\alpha_1 = 1 > 0 = \beta_1$, the only j with $\alpha_j < \beta_j$ is $j = 2$, and $\alpha - e_1 + e_2 = (0, 2, 4)$, whose sorted form $(4, 2, 0)$ does not lie in $P(W)$ because $(4, 2, 0) \not\leq (4, 1, 1)$ and $(4, 2, 0) \not\leq (3, 3, 0)$. The dominance maxima of $P(W)$ are $(4, 1, 1)$ and $(3, 3, 0)$, which are incomparable: their partial sums are $(4, 5, 6)$ and $(3, 6, 6)$. $\square$

6 What the proof consumes

- **Lam, [3, Cor. 8.5]** (affine Stanley $\Rightarrow$ dual k -Schur positivity): *not used*. This is the input Problem 5.1 asks to avoid.
- **Dual k -Schur functions**, k -Kostka numbers, k -Schur positivity, cores and k -bounded partitions: *not used*; they do not appear.
- **Lee 2019, Cor. 5** (cylindric skew Schur = affine Stanley on the 321-avoiding class): *not used*.
- **Postnikov's symmetry theorem** for cylindric skew Schur functions: *not used*.
- **Murota's theory of M-convex sets**: only the definition.
- **Lam [2], symmetry of $\tilde{F}_w$** : *used once*, as Hypothesis 5.1, to obtain Corollary 5.2 and hence S_r -stability. It is used nowhere else; Theorem 4.1 and Corollary 4.4 are independent of it. See §8 for its status in this document.
- **[1, Lem. 4.1 and §6]**, both by the present author and both self-contained: the Robin Hood step for the dominance order, and the passage from an S_r -stable dominance ideal with a maximum to M-convexity.

Everything in §§2–4 is proved here from (1) and the definition of $\tilde{S}_n$.

7 Computational verification

All computations are in Python over the model of §2; code at `code-q209/` in this repository; `verify_all.py` reproduces every number below in one run. The definition (1) is implemented directly, by dynamic programming over chains in the affine weak order, with no use of any theorem below.

Anchors (is the object right?). For $w \in S_n \subset \tilde{S}_n$, $\tilde{F}_w$ must be the ordinary Stanley symmetric function. The coefficient of $x_1 x_2 \cdots x_{\ell(w)}$ in $\tilde{F}_w$ equals the number of reduced words of w , computed independently by recursion on length: agreement in all 106 cases with $n \leq 4$, $\ell(w) \leq 4$. The polynomial $\tilde{F}_w(x_1, \dots, x_r)$ is symmetric in 495 tested pairs (w, r) with $n \in \{3, 4\}$, $\ell(w) \leq 5$, $r \leq 4$ — a nontrivial check of the implementation, and the computational evidence for Hypothesis 5.1.

Block model and Lemma 2.3. The description (2) of u_S agrees with the reduced-word construction for all proper $S \subsetneq \mathbb{Z}/n$, $n \leq 6$. The criterion of Lemma 2.3 agrees with the computed length in 4932 pairs (w, S) with $n \leq 5$, $\ell(w) \leq 4$: 0 disagreements.

Theorem 3.1. The fibre computed from (p, q) agrees with $\mathcal{F}(p^{-1}q)$ in 1106 pairs (p, v) with $n \in \{3, 4\}$: 0 disagreements.

Theorem 4.1. Over $n \in \{3, \dots, 7\}$ and $\ell(v) \leq 5$ (resp. ≤ 4 for $n = 7$) there are 1055 length-additive pairs (S, T) with $|S| > |T|$. The rule of Theorem 4.1 produced a valid smaller factorisation in all 1055 of them. The identity of Lemma 4.3 was exercised 1481 times (once per usable run) and failed 0 times — in particular the rule never produced a pair that was not a length-additive factorisation of v . Step 1 of Lemma 4.2 was checked in 264 instances and its conclusion (a usable run exists) in all 1055: 0 failures.

Variation audit. The fibres are not degenerate. Counting one v per group element with $\mathcal{F}(v) \neq \emptyset$, the distribution of fibre widths $|\mathcal{F}(v)|$ is $\{1 : 11, 2 : 28, 3 : 18, 4 : 4\}$ for $n = 4$, $\ell(v) \leq 5$, and $\{1 : 16, 2 : 25, 3 : 40, 4 : 10, 5 : 5\}$ for $n = 5$, $\ell(v) \leq 4$. Roughly four fifths of fibres have width ≥ 2 , so the interval and palindrome statements are not vacuously true on singletons. Supports $W_r(w)$ of size up to 101 occur. Note that the homogeneity $\sum_t \alpha_t = \ell(w)$ is true by construction and is therefore *not* counted as evidence for anything.

Hypothesis 5.4. Verified, with no counterexample, for $n \leq 5$ ($\ell(w) \leq 7$ for $n = 2$, ≤ 6 for $n = 3$, ≤ 5 for $n = 4$, ≤ 4 for $n = 5$) and $r \leq 5$: 1070 pairs (w, r) with nonempty support, in each of which a single $\alpha \in W_r(w)$ attains $\max\{\alpha_1 + \cdots + \alpha_t\}$ simultaneously for all t . The same computation confirms M-convexity of every $W_r(w)$ in that range, as [4, Thm. 4.7] predicts; this verifies the implementation, not the theorem.

A refuted simplification. We searched for a proof of Theorem 5.5 *without* Hypothesis 5.4. Exhaustively over all subsets of the simplex $\{\alpha \in \mathbb{Z}_{\geq 0}^r : \sum \alpha_i = d\}$ for $(r, d) \in \{(3, 2), (3, 3), (3, 4), (4, 2), (4, 3), (2, 5)\}$, every set with property (2) of Proposition 5.6 is M-convex; the smallest counterexample has $d = 6$ and is the one exhibited there. A search stopping at $d \leq 4$ would have reported the simplification as true.

8 Gaps

Two, stated precisely.

1. **Hypothesis 5.4 is unproved.** This is the whole remaining distance to Problem 5.1. What is wanted: a cyclically decreasing decomposition $w = w^1 \cdots w^r$ maximising $\ell(w^1 \cdots w^t)$ for every t at once. The natural route — that the set A_t of achievable prefixes $w^1 \cdots w^t$ has a maximum in the right weak order, which would give the statement by the induction of [1, Lem. 5.2(3)] — is *false*: over $n \in \{3, 4\}$, $\ell(w) \leq 5$, $r \leq 4$, there are 767 triples (w, r, t) with $|A_t| > 1$, and in 178 of them A_t has no maximum in the right weak order. The smallest is $n = 3$, $w = (1, 0, 5)$, $r = 2$, $t = 1$, where A_1 consists of the two incomparable elements $u_{\{0,1\}}$ and $u_{\{0\}}$. A length maximum still exists there; what is missing is a mechanism forcing it.
2. **Hypothesis 5.1 is cited, not proved here, and the source has not been read at first hand.** It is attributed to [2] and quoted as established at [4, 1.796, 1.802], a source read at first hand. The author has not read [2] itself, so by the author's own source-depth rule no claim resting on it may be graded *proved*. It is used only for Corollary 5.2; Theorem 4.1 and Corollary 4.4 — the new content — do not depend on it. Two ways to close this: read [2] at source, or prove Corollary 5.2 directly. The latter is equivalent to the clean combinatorial statement

$$\min\{|S| : v = u_S u_T \text{ length-additive}\} = \min\{|T| : v = u_S u_T \text{ length-additive}\},$$

which the exchange move alone does not give: it yields only $\min \mathcal{F}(v) \leq \lfloor \ell(v)/2 \rfloor \leq \max \mathcal{F}(v)$.

Neither gap affects Theorem 3.1, Theorem 4.1, Corollary 4.4 or Proposition 5.6, which are proved outright.

References

- [1] Clio. *M-convexity of cylindric skew Schur polynomials*. `projects/proofs/2026-09-20-c1-cylindric-M-convexity.tex`, 20 September 2026.
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- [3] T. Lam. *Schubert polynomials for the affine Grassmannian*. J. Amer. Math. Soc. **21**, 259–281 (2008).
- [4] Wang, Zhang and Zhang. *Newton polytopes and M-convexity of dual k-Schur polynomials*. arXiv:2401.14632.